SPE-227793-MS

Use of Closed-Loop Horizontal Low-Enthalpy Geothermal Well Solution for Seawater Desalination in GCC Countries

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Seawater desalination accounts for 10-20% of the GCC region's energy consumption and 25 % of KSA's annual oil consumption. Instead of using valuable hydrocarbon resources, this paper proposes the use of the abundantly available low-enthalpy geothermal energy, which can be extracted using oil field technology and know-how, through a thermal separation distillation process. A single horizontal closed-loop geothermal well solution, with a vacuumized pipe-in-pipe to nearly eliminate heat loss, can provide sufficient heated water output for desalination with a low risk, environmentally friendly, and virtually without any associated CO_2 emissions.

The solution consists of a vertical well drilled to 3-5 km and with a horizontal section of 2-5 km. Fluid will flow in the outer string in contact with the geological formations, and in this process, the fluid will be heated along the horizontal section to the approximate temperature of the subsurface before returning to the surface in the inner patented vacuumized insulated completion that nearly eliminates any heat loss. As it is a closed-loop solution, there is very limited water consumption, and furthermore, it has no reservoir quality requirements; it can therefore be completed geologically deeper in hotter rocks. Finally, there are no issues with corrosion, scaling, clogging, or downhole maintenance, with the only required maintenance being of a small surface pump to ensure water circulation.

Each geothermal well can produce 1.5-2.5 MWth of heated water, assuming a 150 °C virgin temperature and a sustained 90 °C delivered on the surface. The thermal output depends on the targeted geological formation and the associated thermal conductivity of the horizontal section of the well, which could be Permian/Devonian sandstone or potentially rock salt at a depth of 3-5 km. A cluster of wells is drilled from the same well pad with the horizontal section in a star pattern to reach the required total thermal output.

The most cost-effective method for seawater desalination is to directly provide the abundantly available low-enthalpy geothermal heat to a thermal separation distillation process like a multi-effect distillation plant. This direct use of geothermal energy is more efficient than using electricity. By nearly eliminating the heat loss using the vacuumized pipe-in-pipe solution when producing geothermal energy and having a closed

loop system with limited geological requirements, it is possible to make geothermal energy a key energy pillar in achieving net-zero emission by leveraging the existing oilfield knowledge and expertise and drive new geothermal applications such as desalination using a thermal separation distillation process and district cooling using an absorption chiller.

Introduction

Global environmental and demographic changes are leading to the depletion of natural resources that were once considered abundant. Access to clean water for domestic, industrial, and agricultural purposes has become increasingly limited. Over 70% of the Earth's surface is covered by oceans, but while water is plentiful in total volume, it is largely unavailable in a usable form. Desalination presents a viable pathway to address the scarce freshwater supplies by converting seawater into freshwater.

Low-to-medium enthalpy geothermal resources are increasingly being explored for direct use applications, including water desalination. Geothermal energy is continuous, dispatchable, and has minimal emissions, making it a strong candidate for integration with sustainable desalination systems (Montesdeoca et al., 2023).

The desalination market in Gulf Cooperation Council (GCC) countries is vital for ensuring water security in one of the most water-scarce regions in the world. With significant government investment, technological advancements, and a focus on sustainability, the market is poised for continued growth. However, addressing challenges like energy consumption and environmental impact will be crucial for the long-term success of desalination in the region (Prajapati et al., 2021).

Water desalination is the process of removing salt from saline water (Abdelkareema et al., 2018), and using geothermal heat is an innovative and sustainable approach to producing freshwater by leveraging the Earth's continuous natural heat. Fossil fuels have been the main energy source for the desalination industry, which makes up a significant amount of KSA's annual oil consumption. A single horizontal closed-loop low to medium-enthalpy geothermal well solution, having a vacuumized pipe-in-pipe completion that nearly eliminates heat loss for the fluids delivered at the surface and has a very limited water requirement, can provide sufficient heated fluid output for desalination with low risk, environmentally friendly, and with virtually no CO2 emissions. This solution can deliver 1.5–2.5 MW thermal (MWth) per well, sufficient to drive thermal desalination processes such as a Multi-Effect Distillation (MED) and produce 2,400 to 4,000 m3 freshwater/day.

Geothermal technology

Conventional geothermal systems are based on a doublet design, where one well produces formation fluid and another well injects/returns cooled fluid, with some km's distance apart. The system is therefore highly dependent on the geological formation's thickness and properties such as porosity, permeability, and geochemistry to ensure hydrologic connectivity between the wells to maintain fluid production.

The produced fluids can cause corrosion, scaling, and clogging, resulting in significant maintenance issues in the injector well surface installations and at the reservoir around the injection wellbore. Hydraulic fracking may be required to improve injectivity and formation connectivity, with the risk of compromising groundwater aquifers, having a detrimental impact on the geological formation, and inducing seismicity, which potentially could induce some damage to infrastructure on the surface. To mitigate the issues with the doublet design, a horizontal closed-loop single geothermal well is proposed.

Closed-loop horizontal geothermal well solution

The horizontal closed-loop single geothermal well is drilled to a vertical depth of 3-5 km, depending on the temperature requirement, with a 2-5 km horizontal or slanted section, either as a single well or a group of wells from the same location, depending on the required energy demand (MWth). To complete a closedlooped well, a lower completion liner is installed in the horizontal section, and the vacuumized pipe-in-pipe completion is installed for the total length of the well. The vacuum pump is located on the surface for easy maintenance. The circulation fluid will be heated by flowing down the well between the lower completion liner and the pipe-in-pipe completion in the horizontal section of the well. At the toe of the well, the circulation fluid is returned to the surface in the pipe-in-pipe completion with a minimal heat loss as it works similarly to a thermos flask (Fig. 1). The circulation fluid is "drinking" water that only needs to be supplied once at the start-up of the project (Maver and Rasmussen, 2024).

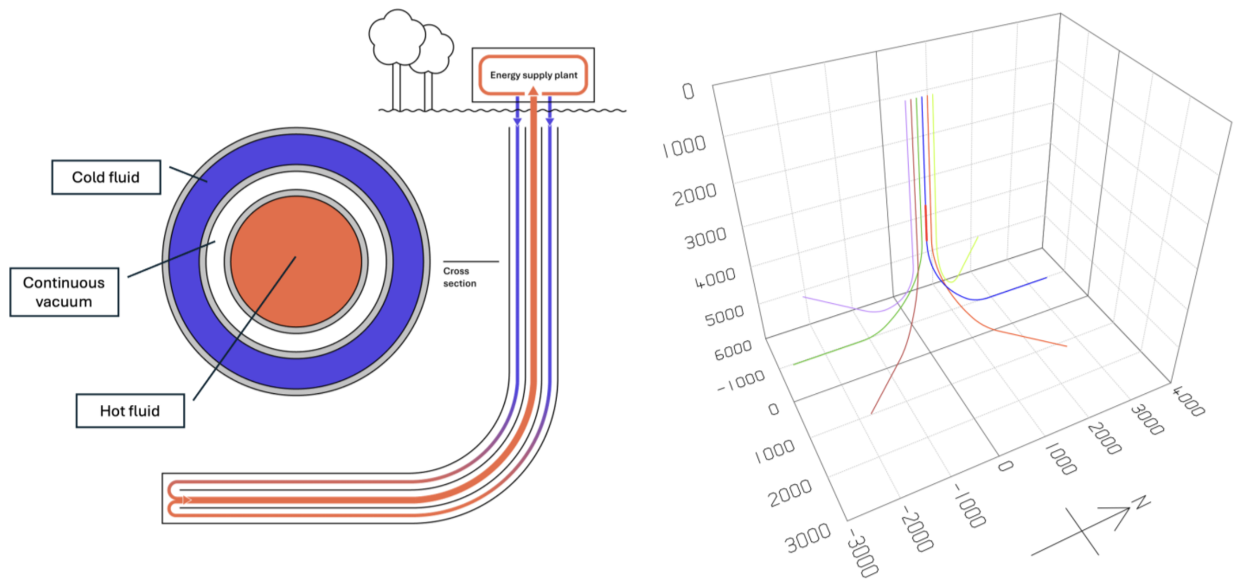

Figure 1—Closed loop geothermal well with the fluid being heated along the horizontal section of the well and returned to the surface at the toe of the well inside the vacuumized pipe-in-pipe that will minimize the heat loss of the returning fluid (Left). A proposed multi-well closed-loop horizontal well design with annotation in meters (Right).

Without the vacuum to insulate the completion, a substantial heat loss can be experienced. Managing this loss of heat is key to making the vacuumized pipe-in-pipe completion unique and making projects commercial. The heat loss has been calculated using a model from Institute for Energy Technology (IFE), Norway, and has been compared and confirmed by various models to a few percent for the full length of the well.

The heat transfer from the geological formation to the circulating fluid in the borehole is key to achieving the required energy output, and it has been modelled using a geothermal simulator developed by the IFE, showing how the temperature and energy output change over time. The model results have also been verified by using several other heat flow models. The output temperature is controlled by the initial temperature of the horizontal section and the fluid circulation speed. The energy output over time is highly dependent on the thermal conductivity of the rock, where the horizontal section is completed, and on the length of the horizontal well section, in a direct relationship.

It is a closed-loop solution making it unnecessary to have downhole equipment and eliminates the issues associated with corrosion, scaling, and clogging. The only required maintenance will be for a small surface pump to ensure fluid circulation. The system is safe with no fracking requirement, potentially inducing seismicity and polluting groundwater aquifers.

The vacuumized pipe-in-pipe completion is designed by the oil service industry by drilling and completion technology specialists who are experienced in drilling deep horizontal wells, completing them with a special production liner with the associated smart well services. To be cost-effective and successfully harvest geothermal energy, the focus is on using oilfield equipment, services, and experience.

Laboratory tests of the vacuumized pipe-in-pipe completion have been completed, as well as extensive field tests completed in the second half of 2024 at the Ullrigg well test site in Stavanger, Norway. Furthermore, at the Gross Schonebeck geothermal test site located approximately 50 km north of Berlin, Germany, the vacuumized pipe-in-pipe completion will be installed in 2025 in a suspended hydrothermal well to a depth of 3,800 m with a bottom-hole temperature of 145 degrees C. Finally, the first full-scale well is expected to be drilled and have the vacuumized pipe-in-pipe completion installed in Denmark in Late 2026.

Completing a single horizontal closed-loop geothermal well

To design the well, the fluid temperature and energy output (MWth) at the surface are the two key parameters, which are controlled by both the initial rock temperature and the thermal conductivity along the length of the horizontal well section. The temperature gradient for the Earth's crust for a deep geothermal well varies and averages 34 °C /km (Thermoglobe, 2025). Areas with, for example, a Precambrian shield have a lower geothermal temperature gradient, and graben structures have a much higher gradient.

The thermal conductivity of the rock has a direct correlation to the produced amount of thermal energy. It varies with the composition of the rock and is controlled primarily by the relative effectiveness of heat transport through grain-to-grain paths of the rock (Robertson, 1988) and usually falls within the range of 0.4–7.0 W/(mk) (Cermak and Rybak, 1982). Low values are characteristic of dry, not consolidated sedimentary rocks, while higher thermal conductivity values are for most sedimentary and metamorphic rocks. Rocks with high quartz content (e.g. quartzite, sandstone) and water-saturated rocks are the best heat conductors, along with rock salt that has a thermal conductivity of slightly less than 5.0 W/(mK) at 160 °C (Raymond et al., 2022). Optimizing the stratigraphic level of the horizontal section of the well is therefore important to achieve the planned energy output (MWth).

Drilling and completing geothermal wells are most common in sedimentary rocks. But as it is a closed-loop system with no requirements for circulating a fluid in the geological formations, the wells can also be completed in both igneous rocks and rock salt. Igneous rocks are more difficult and expensive to drill than sedimentary rocks due to the low Rate Of Penetration (ROP). In contrast, rock salt can be 2-3 times faster than drilling sedimentary rocks, but it is more difficult to run the completion in rock salt.

Multi-well geothermal well solution

It is possible to achieve a 1.5-2.5 MWth energy output per well, depending on the geological setting (thermal conductivity), depth of the horizontal section (temperature), and the length of the horizontal section. The solution is suitable for initial subsurface temperatures around 150 °C or lower, and depending on the flow rate in the well, a long-term temperature of the produced fluids could be 90 °C. Based on the energy requirement, a number of wells can be drilled in a star pattern for a 10-20 MWth total energy output, and the solution can be designed to last 50+ years (**Fig. 1**).

Desalination

Water desalination using geothermal heat harvested from a single closed-loop horizontal geothermal well solution is a low-energy and sustainable approach to producing freshwater.

Technique

Desalination techniques can broadly be categorized into membrane-based and thermal-based methods. Membrane desalination includes technologies such as Reverse Osmosis (RO), ion exchange, and electrodialysis. In contrast, thermal desalination encompasses methods such as MED, Multi-Stage Flash Distillation (MSF), Mechanical Vapor Compression (MVC), and Thermal Vapor Compression (TVC). Regardless of the methods employed, desalination processes are inherently energy intensive. A comparison of widely adopted desalination methods shows that thermal processes generally consume far more energy than membrane-based alternatives, with MSF and especially MED being energy efficient (Prajapati et al., 2021). The majority of the energy consumption for thermal desalination methods is driven by the need of heating the water, however if hot water is supplied directly from a geothermal source in a consistent and sustainable manner, energy use is reduced to only that required by auxiliary equipment.

MED is frequently mentioned as highly compatible with low-to-medium enthalpy geothermal heat sources with typical temperatures below 150 °C, sometimes below 100 °C, or even 70-90 °C (Prajapati et al., 2021, Goosen et al. 2010). Geothermal heat provides the energy to evaporate saline water across multiple stages at decreasing temperatures and pressures (Prajapati et al. 2021), with examples of projects described in Table 1.

Table 1—Isolated cases coupling geothermal with MED technology.

Location/Study	Geothermal Source temperature	MED Configuration	Freshwater Production	Notes
Kimolos Island, Greece	61°C	MED (pilot system)	75m^3/h	Geothermal heat is the sole energy source; offsets oil consumption.
Texas, USA	~150°C	MED	120m^3/day	Production rate varies with geothermal fluid flow.
Western Australia	72°C	Basic/Boosted/Preheated Boosted MED	Not specified	Techno-economic study; several MED design variations evaluated.
Southern Iran (Sim)	Not specified	Seawater MED	Not specified	Numerical model using geo heat from abandoned oil and gas wells.
GCC (Feasibility)	Low enthalpy (exact temp not specified)	MED Vs RO comparison	Not specified	MED explicitly described as being driven by geothermal thermal energy.

Coupling the vacuumized pipe-in-pipe completion directly with a thermal process like MED allows for the direct utilization of the thermal energy delivered at the surface (Fig. 2).

In a coupled system, the fluid is circulated within the vacuumized pipe-in-pipe completion and reaches the surface and is directed toward a MED desalination plant. The integration begins with a heat exchanger positioned as the first effect of the MED unit. This exchanger enables indirect thermal energy transfer from the geothermal fluid to the incoming saline feedwater. It's important that the horizontal well solution remains fully closed and the geothermal fluid does not mix with or come into contact with the seawater or any part of the desalination stream, thereby eliminating the risk of scaling, contamination, or cross-fluid degradation.

The process starts with raw seawater being drawn from the source using feed pumps. The seawater undergoes pre-treatment steps such as filtration and chemical dosing to remove particulates and prevent corrosion or fouling in the distillation stages. Once pre-treated, the seawater is introduced into the first effect of the MED unit, where it flows over the heat exchanger surface. Here, the thermal energy supplied by the 90 °C geothermal fluid is transferred to the saline water, raising its temperature to the point of evaporation, typically around 70 to 75 °C under vacuum conditions.

The resulting water vapor from this first effect then serves as the heating medium for the next stage (second effect), which operates at a slightly lower pressure and temperature. This cascading sequence of evaporation and condensation is repeated across multiple effects, each reusing the heat from the previous stage, thereby significantly increasing the thermal efficiency of the system.

As vapor condenses in each stage, it is collected as freshwater distillate. Meanwhile, the remaining concentrated brine is gradually moved through the stages and ultimately discharged. Throughout this process, internal pumps and control systems manage the flow rates, pressure levels, and vacuum conditions necessary for optimal MED operation with the PFD illustrating the concept of the technology coupling (Fig. 2). After the geothermal fluid has transferred its heat to the first effect, it exits the heat exchanger at a lower temperature and is directed back underground via the well solution. This completes the closed-loop circulation of the vacuumized pipe-in-pipe completion.

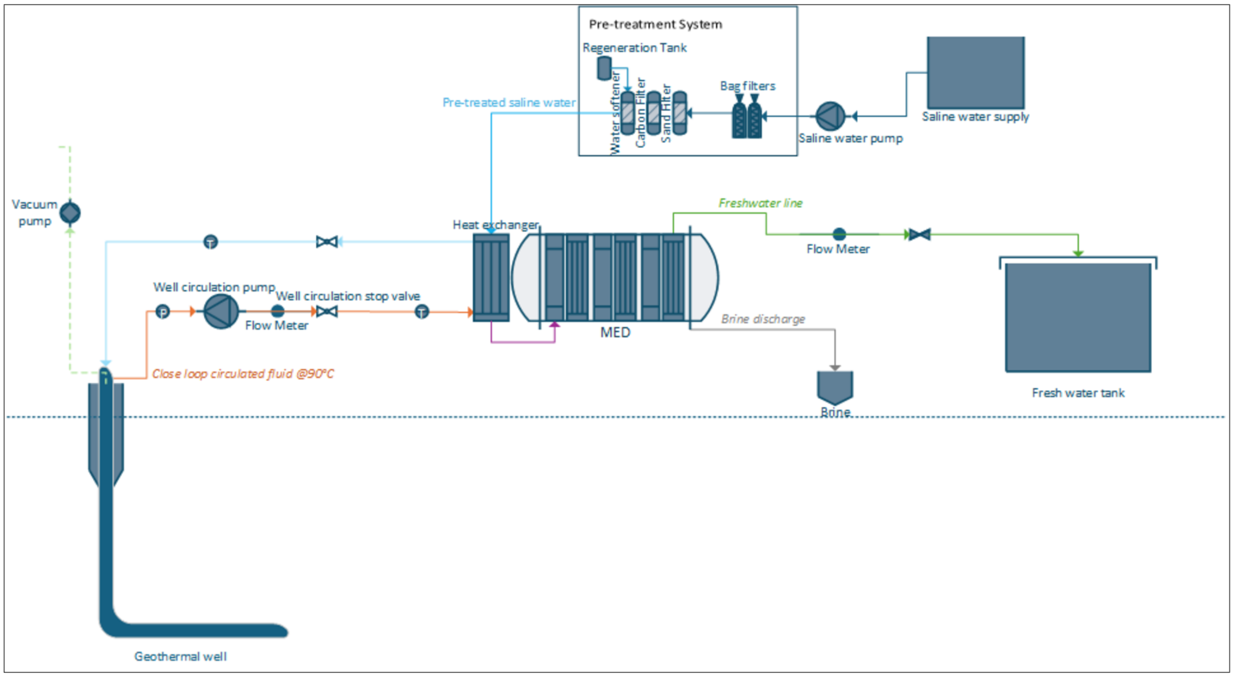

Figure 2—Process flow diagram of the closed loop horizontal geothermal well solution coupled with MED desalination technology.

A MED plant will require external electricity sources to power auxiliary components: Intake pumps, internal circulation pumps, vacuum systems, chemical dosing equipment, and plant control systems. Typically, the electrical energy demand for a MED plant ranges between 2 and 2,5 kWh per cubic meter of freshwater produced (Prajapati et al., 2021).

Freshwater production

To calculate the freshwater production potential of the single closed-loop horizontal well solution when coupled with MED desalination technology, a simple energy balance approach was applied for a geothermal well delivering 1,5-2,5 MWth.

1 MWth is continuously available and fully used by the MED unit for desalination. The equation links the thermal power output of the well to the Specific thermal Energy Consumption (SEC) of the MED system (kWh/m3) of the daily produced freshwater produced:

$$FreshwaterProductionRate\left(m^3/day\right) = \left[\frac{Powerthermaloutput\ (kW)}{SECthermal\ \left(kWh/m^3\right)}\right] \cdot 24$$

According to multiple literature sources, SEC values for MED systems typically range from 12,2 to 19,1 kWh/m^3. An additional dataset from Al Karaghouli and Kazmerski (2013) reports higher SEC values. This variation is stemming from different system boundaries or assumptions in those studies and the more conservative and commonly accepted lower SEC range (12,2-19,1 kWh/m^3) has been used for practical engineering design.

Applying the equation for a 1 MWth source and a median SEC thermal of 15 kWh/m^3, the freshwater production is:

$$1,500\ to\ 2,500\quad kW \div 15\quad kW/m^3 = 100\ to\ 166\quad m^3/h$$

$$\text{or } 2,400\ to\ 4,000\ \mathbf{m^3/day}$$

The range of production capacity numbers found in literature as well as commercial products are consistent with the calculated estimate, supporting the feasibility of producing approximately 2,400 to 4,000 m^3/day from 1,5-2,5 MWth well.

The actual freshwater yield might be lower due to heat transfer losses, auxiliary energy demands, or operational inefficiencies.

Opportunities in the GCC countries

Desalination plays a critical role in the GCC countries, where freshwater scarcity has led to a heavy reliance on seawater desalination technologies. This dependence has historically come at a high energy and environmental cost, as desalination consumes a significant portion of national oil and gas resources. For instance, in KSA, around 25% of petroleum production in 2008 was dedicated to electricity generation and water desalination. Technologies like MSF and RO dominate the region, with MSF accounting for roughly 40% of global desalination plants and being extensively used in the United Arab Emirates, Kuwait, and KSA (Prajapati, et al., 2021). These systems have traditionally been powered by fossil fuels, contributing to high carbon emissions and sustainability concerns.

In recent years, however, there has been a noticeable shift toward integrating renewable energy sources into desalination. KSA is emerging as a leader in this transition, with projects like the Al-Khafji PV-RO plant, which will produce 60,000 m^3/day of freshwater using solar power. Geothermal energy is also gaining attention in the region as a promising alternative. A significant example is a hybrid desalination plant commissioned in a GCC country in 2013 that combines geothermal, photovoltaic (PV), and grid electricity to operate MED and RO technologies, achieving a capacity of 30,000 m^3/day (Prajapati et al., 2021).

Geology

With the single closed-loop horizontal well solution, a geothermal well can be successfully drilled and completed throughout the GCC region. However, there are specific regions and targeted depths that are commercially more attractive than others due to the geothermal temperature gradient, thermal conductivity of the geological formations, and the fast ROP while drilling.

The global average crustal geothermal temperature gradient is 34 °C/km, however, there are significant and very local gradient variations. In Northern Oman, the geothermal gradient is more than 70 °C/km (Al Lamki and Terken, 1996). In KSA, but also applicable for the GCC countries in general, geothermal temperature gradients increase towards the East and South because the depth to the Moho decreases. The temperature gradient in the north-west KSA is around 34°C/km, and the areas around Riyadh have temperature gradients of around 32°C/km (Vahrenkamp et al., 2021).

Arabian Shield consisting of crystalline Precambrian basement with an up to 3 km elevation in Western KSA with hot springs results in a low ROP when drilling transitioning to a thick passive margin sedimentary cover of as much as 10 km thick to the East at the Arabian Gulf enabling a high ROP when drilling (Stern and Johnson, 2010; Vahrenkamp et al., 2021).

There are also opportunities for geothermal energy production by utilizing the high thermal conductivity of salt as well as a high ROP when drilling in significant deep salt deposits in Bahrain, Kuwait, KSA, and UAE at the Arabian Gulf and centrally in Oman (Folle, 2004).

To maximize the economic feasibility of the single closed-loop horizontal well, the region towards the Arabian Gulf and Gulf of Oman is most attractive due to sedimentary thickness, geothermal temperature gradient and in some cases the high thermal conductivity, however as the geothermal temperature gradient is reasonable high, geothermal heat can be produced in the entire region.

Seawater desalination potential

To estimate the thermal energy required to meet the basic freshwater needs of a GCC country using geothermal-powered desalination, a reference population of 1 million is considered. Freshwater consumption rates often exceed 500 liters per person per day in the GCC region, which is significantly higher when compared with figures of countries with comparable income levels, such as Germany, where consumption is around 120 liters per person per day (El-Kogali, 2024). Using the freshwater consumption of Germany, the water requirement per person for a population of 1 million is 120,000 m^3. It has been established that a geothermal well delivering 1,5-2,5 MWth energy to a MED system can produce around 2,400-4,000 m^3 of freshwater per day. This estimate aligns with reported MED energy consumption figures, typically ranging between 12,2 and 19,1 KWh per m^3, and is supported by the literature on geothermal applications for low-enthalpy thermal desalination (Prajapati et al., 2021). Applying this relationship, the thermal capacity required to supply 120,000 m^3/day and an average well of 2 MWth is calculated as:

$$Required\ \ Thermal\ \ power\ \left(MW_{th}\right)=\frac{Total\ \ daily\ \ water\ \ requirements\left(m3/day\right)}{Production\ \ rate\ \ per\ MW_{th}\left(m3/day\ \ per\ MW_{th}\right)}$$

$$Required\ \ Thermal\ \ power\ \left(MW_{th}\right)=\frac{120,000\ \ m3/day}{3,200\left(m3/day\right)/\left(MW_{th}\right)}$$

$$Required\ \ Thermal\ \ power\ \left(MW_{th}\right)=37,5MW_{th}$$

In order to generate 37,5 MWth energy required to cover the recommended minimum water requirements for a population of 1 million, 19 wells are needed, assuming an average production capacity of 2 MWth per well.

Conclusion

By nearly eliminating the heat loss using the vacuumized pipe-in-pipe completion when producing geothermal energy and having a closed-loop system with limited geological requirements, geothermal energy could be key for desalination using a thermal separation desalination process like MED, as it addresses both the environmental and operational challenges of conventional desalination in the GCC region.

One of the key advantages of this approach is its ability to provide a constant and clean heat source without depending on specific underground water reservoirs. This makes it suitable for deployment across a wide range of locations, including inland and coastal areas that may not support traditional geothermal systems.

As it is a closed-loop solution, there is very limited water consumption, and furthermore, it has no reservoir quality requirements; therefore, it can be completed geologically deeper in a hotter rock. There are no issues with corrosion, scaling, or clogging, and therefore no downhole maintenance, with the only required maintenance being of a small surface pump to ensure water circulation, and with the right design, the geothermal well has a lifetime of 50+ years.

The GCC's strong foundation in oil and gas operations can be directly leveraged for the planning and execution of geothermal wells, using existing infrastructure, technology, and skilled professionals.

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